Source SHA-256: `02426d55c1252148f9ec7e518a6f92593988d78cb1c33ecff7e280a8d09fede4`. Repository links identify source paths; newly committed pages may be unavailable online until publication.

Why large-site pollen can estimate regional vegetation

Paper and question. Sugita, S. (2007), "Theory of quantitative reconstruction of vegetation I: pollen from large sites REVEALS regional vegetation composition," *The Holocene* 17(2), 229–241. This is the first Landscape Reconstruction Algorithm (LRA) theory paper, commonly cited as Sugita (2007a). It asks when pollen from large depositional sites can estimate a region's plant-cover composition despite heterogeneous vegetation. The distinction matters here because numerical agreement with a REVEALS program says nothing by itself about whether its inferred cover represents the vegetation around a particular core. Indexed publisher citation and abstract; our source-specific method account.

Scoped conclusion. In Sugita's POLLSCAPE simulations, the publisher abstract reports accurate regional estimates with relatively small standard errors from multiple lakes at least approximately 100–500 ha in size; much smaller lakes can vary strongly between sites. Its target region is 10^4–10^5 km^2. These simulated findings do not measure accuracy at a Nordic core. Our separate author-R/native check covers two later LandClim II example grids. Publisher abstract; two-grid execution.

The modeled quantity and source roles

A pollen percentage divides the counted grains assigned to a taxon by a defined grain sum. It is not that taxon's share of land cover: taxa produce unequal pollen quantities, grains disperse differently, and a basin receives pollen from more than its immediate shore. REVEALS ("Regional Estimates of VEgetation Abundance from Large Sites") uses large-site pollen and taxon-specific relative pollen productivity estimates (PPEs, called RPP in some sources), with dispersal and deposition assumptions, to estimate regional cover fractions. Pollen fall speed, lake or bog type and basin size influence the distance weighting. These input roles are explicit in the later LandClim author-method profile; the accessible 2007 abstract establishes the model's regional question and large-lake/multiple-site conditions but does not expose its equations or parameter tables. Publisher abstract.

```
regional vegetation in a simulated or observed landscape
  → taxon-specific production and transport → large-site pollen counts
  → REVEALS with declared productivity, fall speed and basin assumptions
  → estimated regional taxon-cover composition
```

The original test used POLLSCAPE simulations across landscapes and atmospheric conditions. Its abstract gives no site or taxon count, scenario ledger, or numeric error distribution. Large lakes are the regional evidence unit; the companion LOVE theory uses that regional estimate to address background pollen at small sites. Publisher abstract.

An exact later source-to-result trace

The trace below tests an installed calculation, not a 2007 observation. The released LandClim II author example identifies `WHH05` as a grid-512 lake with a 150 m radius. Its raw source sample at depth 16, 1320 cal BP, contains 301 counted grains across 30 named fields; the author-prepared 1500 BP interval vector has 297 grains across ten positive taxa, with the four-grain difference accounted for by four named, one-grain excluded fields. The age falls inside `[1250,1750)` cal BP; 1500 BP is the interval label, not the sample's measured age. The prepared model uses 31 taxa, and the source parameter table has 31 PPE means and a 31×31 covariance axis. Authenticated source-to-input trace.

The author R release and installed native engine then produce a WHH05 site-level REVEALS estimate for that 1500 BP window. In the retained paired row, *Calluna vulgaris* has author-R mean 0.9248830793

and standard error (SE) 0.0468069771, both fractional-cover units. The row is one modeled taxon fraction and its model SE, not a measured cover fraction. Across the 31 means the composition sums to one to numerical precision. The maximum R/native residual among the window's means is 4.77×10^{-15}, and among SEs 8.90×10^{-16}. WHH05 contributes eight of 25 supported windows in this example. These exact source identities and denominators make a calculation check possible while leaving ecological accuracy untested. WHH05 account; paired rows.

The wider comparison covers 17 prepared lake sites in two grids, 25 bins, 31 taxa, and 184 of 425 possible site-bins with available support. It matches 14,074 declared regional and site-level mean/SE values within the recorded numerical tolerances. The released source is a later LandClim II author example using a 2016 REVEALS implementation; it neither reproduces Sugita's POLLSCAPE scenarios nor proves the 2007 paper's simulated error claim for historical vegetation. Execution report; method profile.

What remains unresolved and how to use this account

The indexed publisher abstract is the direct 2007 evidence available here. Direct full-article access did not yield a lawful text for this account, and the retained source packet examined contains no Sugita 2007 paper PDF or original POLLSCAPE scenario tables. Therefore original numeric scenario errors, standard-error formula, exact atmospheric regimes, taxon list and parameter values are not evaluated. For our later example, the 17 sites are all lakes, so its agreement does not check bog transport; its historical PPE covariance propagation and 1,128-sequence archive provenance have separate limitations. Even a small R/native residual cannot resolve those source and physical-interpretation questions. Source-specific profile; publication boundary.

Readers can inspect the retained WHH05 trace and two-grid result without private input bytes. To verify the two-grid governed execution report from the repository root:

sh

```
PYTHONDONTWRITEBYTECODE=1 PYTHONPATH=packages/bijux-pollenomics-dev/src \
  .venv/bin/python -m bijux_pollenomics_dev.essd_reproduction.example_reexecution \
  --verify-report docs/report/publication-replay/landclim-example-reexecution/report.json
```

A fresh R/native reexecution requires the separately identified lawful author snapshot and pinned R dependency, as specified in the reexecution instructions. It cannot rerun the 2007 simulation.

The unsent source request is for a lawful full Sugita (2007a) paper, its original POLLSCAPE landscape and atmospheric scenario definitions, basin sizes and positions, taxon/PPE/fall-speed inputs, regional truth vectors, REVEALS outputs and error/SE calculations, plus usable source permissions. The unsent qualified-review request is to assess whether a proposed pollen cohort has sufficiently large and appropriately distributed sites, matched taxa and support, and a defensible parameter/uncertainty profile for its target region. No independent scientific endorsement, original figure reproduction or colleague reading outcome is recorded here.